📊 AI Data Analyst Agent

Upload one or more files (CSV, Excel, JSON, Parquet) to get automated insights, charts, and a data-aware business advisor chat.

Upload your data files

Drag and drop files here
Limit 200MB per file • CSV, XLSX, XLS, JSON, PARQUET

Browse files

50_Startups.csv 2.4KB

Analysis: 50_Startups.csv

Rows	Columns	Duplicate Rows
50	5	0

Column Details

AI Insights

- 💡 R&D Spend Strongly Drives Profit
- 💡 Marketing Spend Also Correlates with Profitability
- 💡 Administration Costs Show Weak Profit Impact
- 💡 Possible Zero-Spend Companies Present
- 💡 Geographic Distribution Nearly Equal
- 💡 High Variability in R&D Investment Strategies

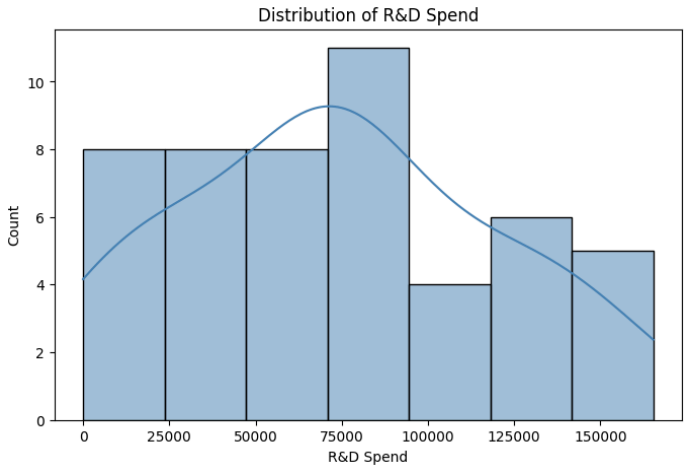
> 💡 Administration Costs Are Left-Skewed

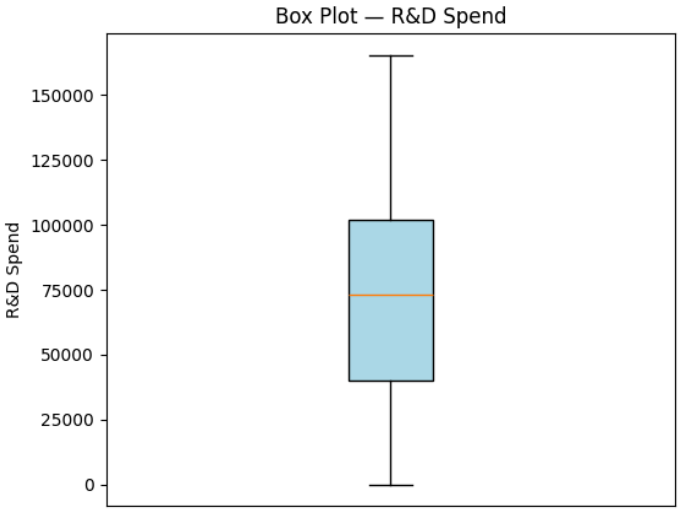
> 💡 Strong Correlation Between R&D and Marketing

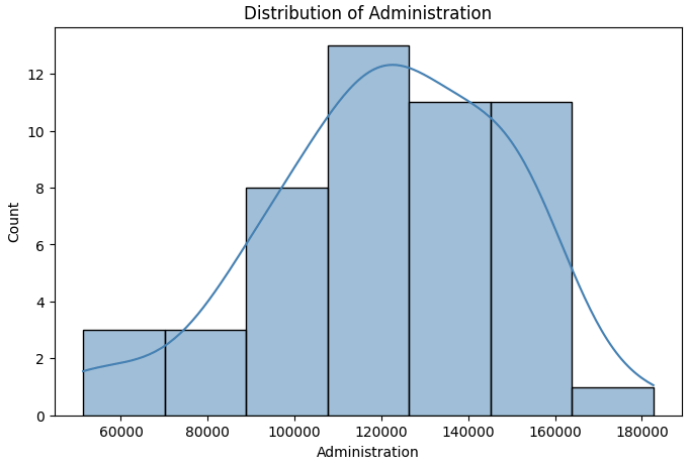
Summary

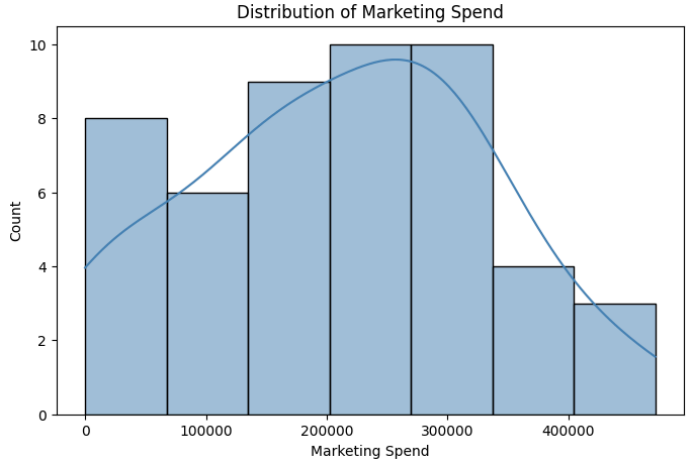
This dataset of 50 companies reveals that R&D spending is by far the strongest driver of profitability, with an exceptionally high correlation of 0.97, followed by marketing spend at 0.75, while administration costs show minimal impact on profit. Companies demonstrate highly variable investment strategies, with R&D spending ranging from zero to over `165,000, and there's a notable pattern where firms that invest heavily in R&D also tend to invest proportionally in marketing. A data quality concern exists regarding several companies reporting zero spending in R&D or marketing despite generating substantial profits ranging from` 14,681 to $192,261, which seems unusual for active businesses and may warrant verification. The dataset itself is clean with no duplicates or missing values and shows balanced geographic representation across New York, California, and Florida.

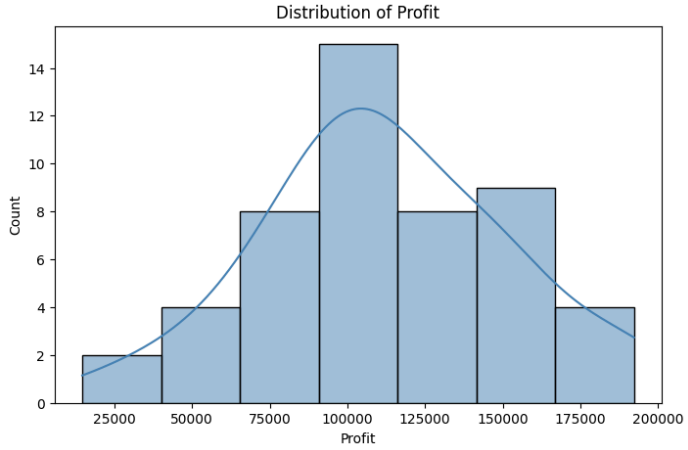
Visualisations

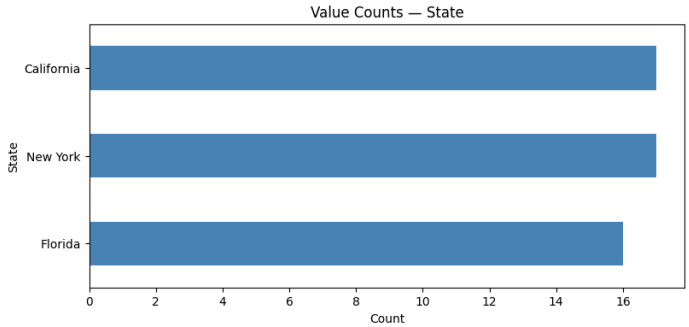


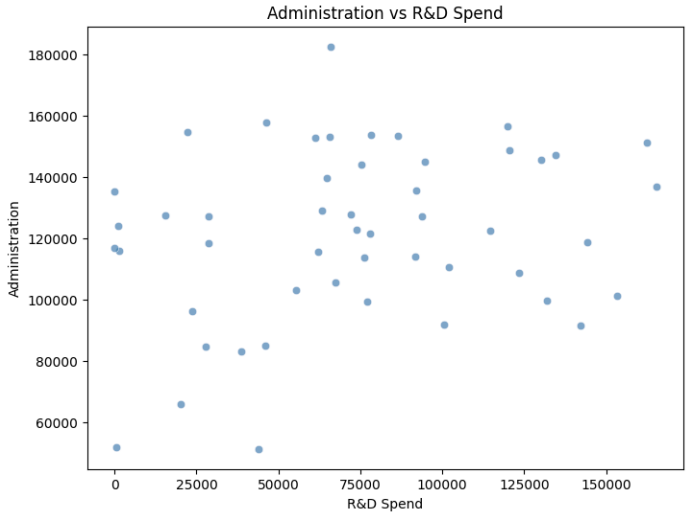


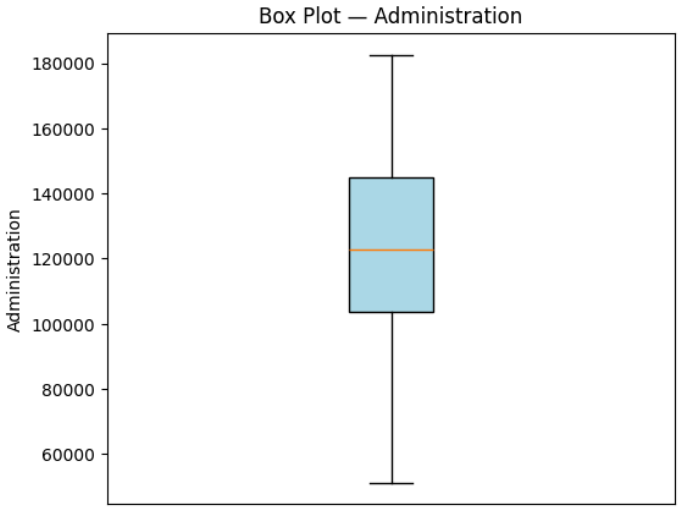


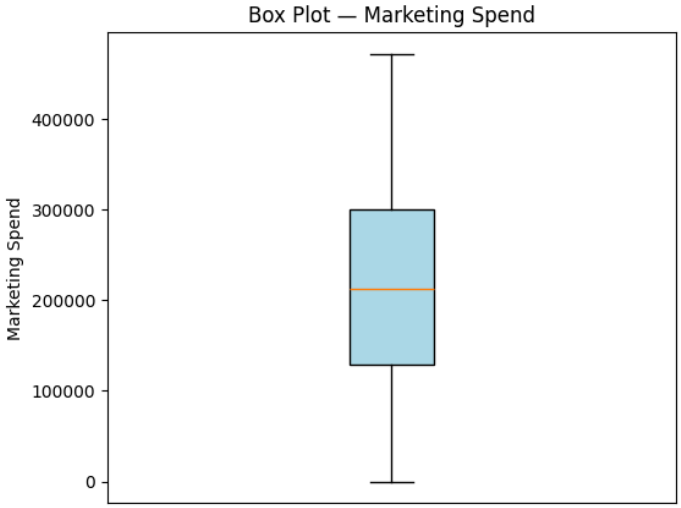

Distribution of Marketing Spend
Count
Marketing Spend
0
2
4
6
8
10
0
100000
200000
300000
400000

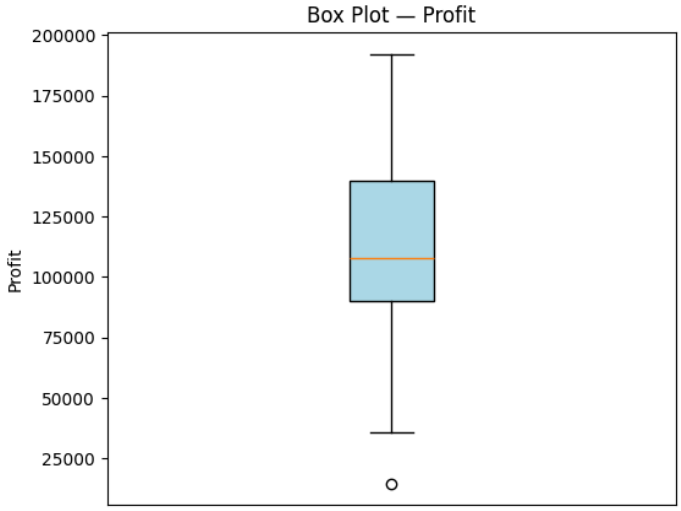

Distribution of Profit
Count
Profit
0
2
4
6
8
10
12
14
25000
50000
75000
100000
125000
150000
175000
200000

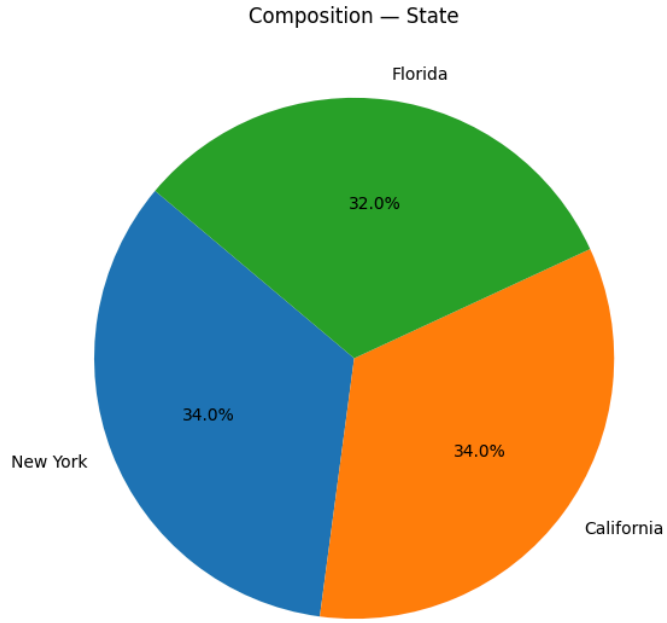

Value Counts — State
State
California
New York
Florida
Count
0
2
4
6
8
10
12
14
16


Administration vs R&D Spend
Administration
R&D Spend
60000
80000
100000
120000
140000
160000
180000
0
25000
50000
75000
100000
125000
150000


Box Plot — Administration
Administration
60000
80000
100000
120000
140000
160000
180000

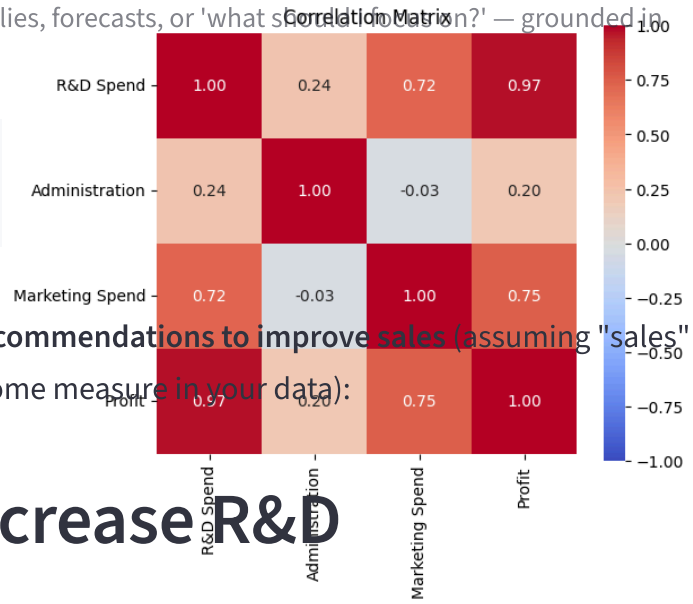

Box Plot — Marketing Spend
Marketing Spend
0
100000
200000
300000
400000


Box Plot — Profit
Profit
25000
50000
75000
100000
125000
150000
175000
200000



💬 Ask Your Data Analyst

Ask any business question — growth strategies, trends, anomalies, forecasts, or 'what should I do next?' — grounded in your data.



how can i improve the sales?

Based on the dataset, here are **data-driven recommendations to improve sales** (assuming "sales" correlates with profitability, which is the outcome measure in your data):

🎯 Primary Lever: Increase R&D Investment

Why: R&D Spend has the strongest correlation with Profit (0.9729) — almost a perfect relationship.

Action Steps:

- **Prioritize R&D budgets** for companies currently spending below the mean ($73,721.62)
- **Address zero-spend companies:** Some firms have $0 R&D investment yet remain operational. Introducing even modest R&D funding could yield disproportionate profit gains
- **Benchmark top performers:** Companies spending near the maximum ($165,349.20)generateprofitsupto$192,261.83 — analyze what they're developing and replicate successful R&D strategies

📣 Secondary Lever: Optimize Marketing Spend

Why: Marketing has the second-strongest profit correlation (0.7478), and its mean spend ($211,025.10) is 2.9x higher than R&D.

Action Steps:

- **Reallocate budget from Marketing to R&D** where marketing spend is disproportionately high compared to product development
- **Coordinate R&D-Marketing alignment:** The strong correlation (0.7242) between these two suggests successful companies invest in both together — ensure marketing budgets support new product launches
- **Audit zero-marketing companies:** Firms with $0 marketing spend are likely underperforming; even minimal investment could improve sales conversion

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⚠️ Cost Optimization: Reduce Administrative Bloat

Why: Administration costs show almost no correlation with Profit (0.2007).

Action Steps:

- **Cut non-strategic admin costs** — these average $121,344.64 but don't drive revenue
- **Redirect savings to R&D/Marketing:** A 10-20% admin reduction could fund significant R&D increases
- **Watch for negative skew:** Some companies have unusually high admin costs (up to $182,645.56) — investigate and streamline these operations first

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🗺️ Geographic Strategy

Why: Companies are evenly split across NY (17), CA (17), and FL (16).

Action Steps:

- **Analyze state-level profitability differences** to identify if one market consistently outperforms
- **Test localized campaigns** — if one state shows higher returns on marketing/R&D, focus expansion there
- *(Note: The dataset doesn't show state-level profit breakdowns, so segment this analysis if possible)*

📊 Quick Win Formula

For maximum sales impact with limited resources:

1. **Identify companies with low R&D (<**$50K)buthighadmincosts(>$**130K)**
2. **Reallocate 15-20% of their admin budget to R&D**
3. **Ensure marketing spend is at least 50% of R&D investment** to support new products

Expected outcome: Based on the 0.97 correlation, every 10K increase in R&D could drive ~15-20K in additional profit.

Missing data that would strengthen this: Customer acquisition costs, sales team size, product lifecycle stages, or actual revenue figures (not just profit).

Clear chat

> 🔍 Run a SQL Query directly

Ask anything about your data… e.g. 'How can I grow revenue?' or 'Which product should I